

Manual

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1 Flatcam:

1.1 calculate tool diameter

tool→calculator input tip-Diameter as 0.2mm input tip angel as 60 input desired cut-z as 0.1 use the tool diameter given in isolating routing

1.2 Isolation Routing

File→Open→Open Gerber in Project tab select the gerber file that has your isolation routing select isolation routing under Tools in gerber object isolation In isolation tool tool add from DB and input the tool Dia that was calculated in the tool calculator

press the generate geometry button

in parameters for Tool 1, select the dia of the v-tip and input 0,02 then input the angle of the v-tip 60, both both x-y and z feed rates between 0-170 (170 recommended) set the spindle speed at 11000. press Generate CNCJob object save the cnc code for the usb drive.

1.3 Drilling

File→Open→Open Excellon select drilling Tool

select the tools you want to create the CNC job fore

Adjust Drill Z to the thickness of the copper laminate(value must be negative)

set travel Z to 2.00 mm

set feed rate between 0 and 170(170 recommended) set spindle speed to 11000.

click Create CNC Job object

save the cnc code for the usb drive

1.4 Board Cutout

File→→OpenOpen Gerber for your board cutout select your board outline Under Tools select cutout tool specify a Margin. This will create a rectangular cutout at the given distance from any element in the Gerber. Specify a Gap Size. 2 times the diameter of the tool you will use for cutting is a good

size. Specify how many and where you want the Gaps along the edge, 2 (top and bottom), 2 (left and right) or 4, one on each side.

Click on Generate Geometry.

Create a CNC job for the newly created geometry

1.5 2-side PCB

open File→Open→Open Gerber Select the side that has not been used (copper top/bottom)

select Tools→2-sided PCB tool under type select gerber and the side of the pcb you are working on Then at reference select box and under reference object select gerber and board outline (depends on file name) and then select mirror.

Select drill dia (usually 1 or 1.5mm). Select alignment drill coordinates by right click at the coordinate then press + and Generate excellon object.

Go to Project select the gerber file that was mirrored proceed as in isolation routing

Go to Project select the alignment hole file (name may differ) proceed as in drilling

2 Wegster operation:

2.1 start up

load the g-code files that you wish to use onto the computer in a way that you can find them. Open Wegstr CNC controlling software if it is not already open. if the indicator under status does not read device ready push the button next to start on the wegstr CNCMilling Machine clamp down the copper laminate to the machine bed locate the bits with corresponding dimensions as the tools used in FlatCAM

2.2 safety

The manufacturer recommends the use of eye protection and protective gloves while operating the machine.

2.3 Isolation Routing

change the bit to the spiral bit press g-code to load files. pick the isolation routing file. Use the X+- y+- z+- button to get the origin then press zero all button in the software left of pause and stop. press the button in the software under auto level. than press the button labeled start use the vacuum to remove the fiberglass recommended to use while the Routing is tacking place

2.4 Drilling

If you start with drilling press autolevel when the drill is touching the copper after auto lever has finished press start. If you started with isolation routing Place the drill outside of the isolation routing with the x,y,z buttons and drag it down to the copper make sure when it is touching the copper zero Z. Then press start. Repeat for all drill files if you have multiple make sure to use the correct drill bits.

2.5 Board Cutout

this is the last operation for the PCB change the bit to the Flat end milling bit load the g-code for the board cutout Take the milling bit outside of the

pcb board onto the copper plate push down on the z axis until it touches the copper plate 0 the z axis and start milling the outline of the board. press start use the vacuum to remove the fiberglass recommended to use while the milling is tacking place

2.6 2-side PCB

load the alignment drill g-code change the bit to your drill bit Take the drill to a spot outside of the pcb make sure to push down z axis to the copper plate and then 0 the z axis then start drilling the allignment holes. Make sure when flipping the plate that the is pushed down in the hole and has the same axis parameters as the opposite side while in the hole. clamp the laminate down to the work bed load the isolation routing g-code fore the side not used change to the spiral bit press auto level press start button

3 Links:

3.1 flatcam:

<http://flatcam.org/manual/index.html>

3.2 wegstr manual pdf:

<https://testglassware.com/wp-content/uploads/2022/05/MANUAL-ITEM-1.pdf>